MODERN PHYSICS – MIDTERM 2.

Please write every problem on separate sheet of paper, and write your Neptun code and Name on every paper you submit. Make sure to write all the formulas that you evaluate, as well as your calculations, so you can get partial score even if there is an error. Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. Note that if a problem is scored, e.g., X + Y + Z points, then *three* answers are expected. Read and start every part of the problems, you can still get partial score even if you don't have the earlier results.

Physical constants and useful formulas

$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$ $\quad h = 6.626 \times 10^{-34}\,\mathrm{J\,s} = 4.136 \times 10^{-15}\,\mathrm{eV\,s}$ $\quad c = 3.00 \times 10^{8}\,\mathrm{m/s}$

$m_e c^2 = 511\,\mathrm{keV}$ $\quad hc = 1.986 \times 10^{-25}\,\mathrm{J\,m} = 1240\,\mathrm{eV\,nm}$ $\quad \frac{\hbar^2}{2m} = 3.810 \times 10^{-2}\,\mathrm{eV nm^2}$

$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$ $\quad f' = f\sqrt{\frac{1-v/c}{1+v/c}}$ $\quad u' = \frac{u-v}{1-\frac{uv}{c^2}}$

1. Doppler effect (3 + 4 = 7 points)

A space probe is traveling towards the moon at a speed of $v_1 = 0.01\,c$ sends back information to Earth at a rate of $f_0 = 39\,\mathrm{MHz}$.

(a) At what rate f_1 does Earth receive the information?

The probe utilizes a lunar gravity assist (or *slingshot*) to alter its trajectory to return to Earth.

(b) What is its velocity v_2, if Earth now receives the information at a rate $f_2 = 39.9\,\mathrm{MHz} > f_0$?

2. Leptonic decay $\phi \to K^- K^+$ (4 + 5 + 2 + 2 = 13 points)

A ϕ meson with rest mass $m_\phi c^2 = 1.02\,\mathrm{GeV}$ is traveling at a speed of $v = 0.50\,c$ in the lab frame when it decays into a pair of kaons K^- and K^+, both with rest mass $m_K c^2 = 494\,\mathrm{MeV}$.

(a) What is the *total* kinetic energy T of the two kaons in the *rest frame* of the decaying meson?

(b) What is the speed of the kaons u' in the *rest frame* of the decaying meson?

(c) What are (i) the highest speed $u_{\max}$ and (ii) the lowest speed $u_{\min}$ the kaons may have in the *lab frame*?

3. Fossil dating with ^{40}K (5 points)

Potassium-argon (K-Ar) dating is a radiometric method used in geology and archaeology to determine the age of rocks and minerals by measuring the decay of the radioactive ^{40}K isotope into the stable ^{40}Ar. It was found that the ^{40}K isotope in a rock sample surrounding the dinosaur fossil of a *Tyrannosaurus Rex* had an activity of $A = 9.64\,\mathrm{decays/minute}$. In contrast, a freshly formed rock sample of the same composition would exhibit an activity of $A_0 = 10\,\mathrm{decays/minute}$. Assume that the two samples contain the same amount of ^{40}K at the time of their formation. Knowing that the half-life of the ^{40}K isotope is $T_{1/2} = 1.25 \times 10^9\,\mathrm{y}$, estimate the age of the fossil.

Comment: for these kinds of problems, it is convenient to express the result in years [y] using scientific notation, and/or millions of years [My], rather than in the usual SI unit [s].

4. Stern-Gerlach experiment with a twist (2 + 4 + 3 + 2 = 11 points)
In a Stern-Gerlach experiment, silver atoms are passed through a region with magnetic field gradient in the $\hat{n}$ direction, where $\hat{n}$ is a normalized vector pointing 45° from the x axis in the xy plane. Now the silver atoms are split according to their eigenvalues with respect to the operator

$$S_{\hat{n}} = \hat{n} \cdot \vec{S} = \frac{1}{\sqrt{2}}(S_x + S_y),$$

where

$$\hat{S}_x = \frac{\hbar}{2}\hat{\sigma}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2}\hat{\sigma}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(a) Show that the normalized eigenstate of $S_{\hat{n}}$ with $+\hbar/2$ eigenvalue is

$$|+\rangle = \frac{1}{2}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix}.$$

(Check that it is an eigenvector, and that it is normalized.)

(b) Find the other normalized eigenstate $|-\rangle$ with $-\hbar/2$ eigenvalue.

(c) Calculate the expectation values of $\langle +| S_x |+\rangle$, $\langle +| S_y |+\rangle$, and $\langle +| S_z |+\rangle$.
Hint: Check that your result is consistent with the classical picture that the spin is polarized in the $\hat{n}$ direction.

(d) We direct the beam of electrons with the $|+\rangle$ state into another Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the two beams observed after the second setup relative to the intensity of the incoming beam?
Hint: Calculate the probabilities $P(+z)$ and $P(-z)$ of the electrons in the $|+\rangle$ state pointing in the $+z$ and $-z$ direction. Remember that $P(+z) + P(-z) = 1$.

5. Variational Methods: Pushed against a wall (2 + 4 + 4 + 4 = 14 points)
Consider an electron in one dimension, with an infinite potential barrier blocking the left half-line, and a constant force F pushing the electron against this wall. It is described by the Hamiltonian $H = K + V$, where $K = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$ is the kinetic term and the potential is given by

$$V(x) = \begin{cases} \infty & \text{if } x < 0, \\ Fx & \text{if } 0 \le x, \end{cases}$$

where $F = 1\,\text{eV/nm}$. Consider a variational wavefunction, which we choose to satisfy Dirichlet boundary condition at $x = 0$, characterized by a single parameter a with nm units, and a normalization constant N:

$$\psi(x) = Nxe^{-x/a} \quad \text{for } 0 \le x.$$

Hint: The electron can only move in the right half-line with $0 \le x$. Use the integral identity $\int_0^\infty x^n e^{-x/\alpha} dx = n!\,\alpha^{n+1}$.

(a) Write the normalization condition and show that it is fulfilled for $N = \frac{2}{a^{3/2}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K \rangle_\psi$ that depends parametrically on a. (Don't forget the normalization constant!)

(c) Calculate the expression for the potential energy expectation value $\langle V \rangle_\psi$ that depends parametrically on a.

(d) Calculate the expression for the energy expectation value $E(a) = \langle H \rangle_\psi = \langle K \rangle_\psi + \langle V \rangle_\psi$, and apply the variational principle to find an equation for a_0 in the ground state. Calculate the approximate ground state energy $E_0 = E(a_0)$ in electronvolt units to four significant digits.
Hint: as a reference value, the exact solution for this potential is $E_0 \approx 0.7867\,\text{eV}$.

OFFICIAL SOLUTIONS

1. Doppler effect (3 + 4 = 7 points)
A space probe is traveling towards the moon at a speed of $v_1 = 0.01\,c$ sends back information to Earth at a rate of $f_0 = 39\,\mathrm{MHz}$.

(a) At what rate f_1 do Earth receive the information?

The probe utilizes a lunar gravity assist (or *slingshot*) maneuver to alter its trajectory to return to Earth.

(b) What is its velocity v_2, if Earth now receives the information at a rate $f_2 = 39.9\,\mathrm{MHz} > f_0$?

Solution
The key is that the probe velocity changes sign due to the slingshot maneuver. If we follow the convention that the velocity is *negative/positive* for a receding/approaching source, the relativistic Doppler formula is given by

$$f = f_0 \sqrt{\frac{1+\beta}{1-\beta}}.$$

Then

(a) When the probe is receding from Earth ($\beta = -v_1/c < 0$) the rate of the signal received on Earth is given by

$$f_1 = f_0 \sqrt{\frac{1-v_1/c}{1+v_1/c}} \simeq 38.6\,\mathrm{MHz} < f_0 \ \text{(redshift)}.$$

(b) After the slingshot maneuver, the probe is approaching Earth ($\beta = v_2/c$) so that

$$f_2 = f_0 \sqrt{\frac{1+v_2/c}{1-v_2/c}} > f_0 \ \text{(blueshift)}.$$

Solving for v_2 yields

$$v_2 = \left(\frac{f_2^2 - f_0^2}{f_2^2 + f_0^2}\right) c \simeq 0.0228\,c$$

2. Leptonic decay $\phi \to K^- K^+$ (4 + 5 + 2 + 2 = 13 points)
A ϕ meson with rest mass $m_\phi c^2 = 1.02\,\mathrm{GeV}$ is traveling at a speed of $v = 0.50\,c$ in the lab frame when it decays into a pair of kaons K^- and K^+, both with rest mass $m_K c^2 = 494\,\mathrm{MeV}$.

(a) What is the *total* kinetic energy K of the two kaons in the *rest frame* of the decaying meson?

(b) What is the speed of the kaon u' in the *rest frame* of the decaying meson?

(c) What are (i) the highest speed $u_{\max}$ and (ii) the lowest speeds $u_{\min}$ the kaons may have in the *lab frame*?

Solution

(a) In the frame of the decaying ϕ meson, the total energy is

$$E = E_{0\phi} = 1.02\,\mathrm{GeV}$$

and since the energy is conserved in the decay process

$$E = K + 2E_{0K}$$

so that the difference of rest masses is transformed into kinetic energy available for the kaon pair

$$K = E_{0\phi} - 2\,E_{0K} = 32.0\,\mathrm{MeV}$$

(b) In that reference frame, the K^- and K^+ particles must recoil with equal momenta in opposite directions in order to conserve momentum. Hence, the kinetic energy of each kaon is

$$K_K = K/2 = 16.0\,\mathrm{MeV}.$$

The speed of the kaon can be found by calculating γ

$$\gamma = \frac{E_{0K} + K_K}{E_{0K}} \simeq 1.03,$$

so that

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} \simeq 0.249 \qquad \text{or} \qquad u' \simeq 0.249\,c.$$

Note that the symbol u' was used to emphasize that the velocity is obtained in the reference frame co-moving with the meson.

(c) The highest and lowest speeds in the lab frame are obtained when the kaons are released in the forward and backward directions with respect to the moving meson, respectively. Then, by using the velocity addition laws

$$u_{\max} = \frac{u' + v}{1 + \dfrac{v}{c^2}u'} \simeq 0.666\,c \qquad \text{and} \qquad u_{\min} = \frac{u' - v}{1 - \dfrac{v}{c^2}u'} \simeq 0.287\,c.$$

3. Fossil dating with ^{40}K (5 points)
Potassium-argon (K-Ar) dating is a radiometric method used in geology and archaeology to determine the age of rocks and minerals by measuring the decay of the radioactive ^{40}K isotope into the stable ^{40}Ar. It was found that the ^{40}K isotope in a rock sample surrounding the dinosaur fossil of a *Tyrannosaurus Rex* had an activity of $A = 9.64\,\text{decays/minute}$. In contrast, a freshly formed rock sample containing the same amount of ^{40}K would exhibit an activity of $A_0 = 10\,\text{decays/minute}$. Knowing that the half-life of the ^{40}K isotope is $T_{1/2} = 1.25 \times 10^9\,\text{y}$, estimate the age of the fossil.

Comment: for these kinds of problems, it is convenient to express the result in years [y] using scientific notation, and/or millions of years [My], rather than in the usual SI unit [s].

Solution
Radioactivity follows the decay law

$$N = N_0 e^{-\lambda t} = N_0 e^{-\ln(2)t/T_{1/2}}$$

where, in the case of ^{40}K, the decay constant is given by $\lambda = \ln(2)/T_{1/2} \simeq 5.55 \times 10^{-10}\,\text{y}^{-1}$.

The activity is defined as

$$A = -\frac{dN}{dt} = \lambda N.$$

The ratio of the activities then gives

$$\frac{A}{A_0} = \frac{N}{N_0} = e^{-\lambda t}$$

so that

$$t = -\ln\left(\frac{A}{A_0}\right)\frac{1}{\lambda} \simeq 6.60 \times 10^7\,\text{y}.$$

From the measurement performed, the fossil was determined to be circa 66 million years old, placing it at the end of the Cretaceous period.

Curiosity: ^{40}K has two decay channel: $^{40}\text{K} \to {}^{40}\text{Ca}$ (89%) and $^{40}\text{K} \to {}^{40}\text{Ar}$ (11%). The problem formulated in terms of activities already takes into account the different decay channel probability. However, conventionally the amount of ^{40}K and ^{40}Ar remaining in the sample is measured, and one needs to corrects for unmeasured fraction of ^{40}K which decayed into ^{40}Ca.
Despite ^{40}Ar is the less common decay product, it is inconvenient to measure the content of ^{40}Ca in the sample because the tiny amount of radiogenic calcium is completely swallowed up by the massive amount that is naturally occurring in Earth's crust. Instead, being ^{40}Ar an inert noble gas, it bubbles out and escapes into the atmosphere when the rock is in a hot, molten state, but is permanently trapped in the crystal, once the rock cools and solidify.

4. Stern-Gerlach experiment with a twist (2 + 4 + 3 + 2 = 11 points)
In the Stern-Gerlach experiment, a beam of neutral silver atoms ($S = J = 1/2$, $L = 0$) are passed through a magnetic field gradient in the $\hat{n} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ direction ($\hat{n}$ is a normalized vector pointing 45° from the x axis in the xy plane). Now the silver atoms are split according to their eigenvalues with respect to the operator

$$S_{\hat{n}} = \hat{n} \cdot \vec{S} = \frac{1}{\sqrt{2}}(S_x + S_y),$$

where

$$\hat{S}_x = \frac{\hbar}{2}\hat{\sigma}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2}\hat{\sigma}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(a) Show that the normalized eigenstate with $+\hbar/2$ eigenvalue is

$$|+\rangle = \frac{1}{2}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix}.$$

(Check that it is an eigenvector, and that it is normalized.)

(b) Find the other normalized eigenstate $|-\rangle$ with $-\hbar/2$ eigenvalue.

(c) Calculate the expectation values of $\langle +| S_x |+\rangle$, $\langle +| S_y |+\rangle$, and $\langle +| S_z |+\rangle$.
Hint: Check that your result is consistent with the classical picture that the spin is polarized in the $\hat{n}$ direction.

(d) We direct the beam with $+\hat{n}$ polarization ($|+\rangle$ with $+\hbar/2$ eigenvalue) into another Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the two beams observed after the second setup relative to the intensity of the incoming beam?

Solution

(a) It can be easily verified that

$$\frac{1}{\sqrt{2}}(S_x + S_y)|+\rangle = \frac{\hbar}{4\sqrt{2}}\begin{pmatrix} 0 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{4\sqrt{2}}\begin{pmatrix} 2 \\ \sqrt{2}(1+i) \end{pmatrix} = \frac{\hbar}{2}|+\rangle.$$

and

$$\langle +|+\rangle = \frac{1}{4}\left(\sqrt{2},\ 1-i\right)\cdot\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{1}{4}(2+2) = 1.$$

(b) We rescale the Hamiltonian by removing the $\hbar/2$ prefactor, and look for an eigenvector with -1 eigenvalue in the form

$$|-\rangle = \begin{pmatrix} u \\ v \end{pmatrix}.$$

The eigenvalue-eigenvector equation reads

$$\frac{1}{\sqrt{2}}(S_x + S_y)|-\rangle = -\frac{\hbar}{2}|-\rangle, \tag{1}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = -\begin{pmatrix} u \\ v \end{pmatrix}, \tag{2}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} (1-i)v \\ (1+i)u \end{pmatrix} = -\begin{pmatrix} u \\ v \end{pmatrix}.$$

From the second line we get $v = -\frac{1}{\sqrt{2}(1+i)}u$. Choosing $u = \sqrt{2}$, this results in the unnormalized eigenvector

$$|-\rangle = N \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix}.$$

The normalization condition reads

$$1 = \langle -|- \rangle = N^2 \begin{pmatrix} \sqrt{2}, & -(1-i) \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = N^2(2+2) = 4N^2.$$

So $N = 1/2$, and the normalized eigenvector is

$$|-\rangle = \frac{1}{2} \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix}.$$

(c) The expectation values are given by

$$\langle +| S_x |+\rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1+i \\ \sqrt{2} \end{pmatrix} = \frac{\hbar}{8}(\sqrt{2}(1+i) + \sqrt{2}(1-i)) = \frac{\hbar}{2\sqrt{2}},$$

$$\langle +| S_y |+\rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1-i \\ i\sqrt{2} \end{pmatrix} = \frac{\hbar}{8}(\sqrt{2}(1-i) + \sqrt{2}(1+i)) = \frac{\hbar}{2\sqrt{2}},$$

$$\langle +| S_z |+\rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix} = \frac{\hbar}{8}(2-2) = 0.$$

This result is consistent with the classical picture, that the spin vector has magnitude $\hbar/2$ and points in the direction of $\hat{n}$, and the spin expectation values are the same as the projections of this vector.

(d) The intensities of the two beams after the setup with z orientation are proportional to the probabilities of the spins pointing in the $\pm z$ direction. This probability is given by the squared magnitudes of components of the spin state given in the standard S_z basis (where the S_z operator is diagonal). Hence the probabilities in state $|+\rangle$ are:

$$P(+z) = \left| \frac{\sqrt{2}}{2} \right|^2 = \frac{1}{2},$$

$$P(-z) = \left| \frac{1+i}{2} \right|^2 = \frac{1}{2}.$$

So the relative intensities of the two beams are 1/2 and 1/2 of the incoming beam.

4. Variational Methods: Pushed against a wall (2 + 4 + 4 + 4 = 14 points)
Consider an electron in one dimension, with an infinite potential barrier blocking the left half-line, and a constant force F pushing the electron against this wall. It is described by the Hamiltonian $H = K + V$, where $K = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$ is the kinetic term and the potential is given by

$$V(x) = \begin{cases} \infty & \text{if } x < 0, \\ Fx & \text{if } 0 \le x, \end{cases}$$

where $F = 1\,\mathrm{eV/nm}$. Consider a variational wavefunction, which we choose to satisfy Dirichlet boundary condition at $x = 0$, characterized by a single parameter a with nm units, and a normalization constant N:

$$\psi(x) = Nxe^{-x/a} \quad \text{for } 0 \le x.$$

Hint: The electron can only move in the right half-line with $0 \le x$. Use the integral identity $\int_0^\infty x^n e^{-x/\alpha}dx = n!\,\alpha^{n+1}$.

(a) Write the normalization condition and show that it is fulfilled for $N = \frac{2}{a^{3/2}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K\rangle_\psi$ that depends parametrically on a. (Don't forget the normalization constant!)

(c) Calculate the expression for the potential energy expectation value $\langle V\rangle_\psi$ that depends parametrically on a.

(d) Calculate the expression for the energy expectation value $E(a) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi$, and apply the variational principle to find an equation for a_0 in the ground state. Calculate the approximate ground state energy $E_0 = E(a_0)$ to four significant digits.
Hint: as a reference value, the exact solution for this potential is $E_0 \approx 0.7867\,\mathrm{eV}$.

Solution

(a) We write the normalization condition

$$1 = N^2\int_0^\infty x^2e^{-2x/a}dx = N^2 2\left(\frac{a}{2}\right)^3 = N^2\frac{a^3}{4} \quad \Rightarrow \quad N = \frac{2}{a^{3/2}}.$$

(b) We write the kinetic energy expectation value

$$\langle K\rangle_\psi = \frac{4}{a^3}\int_0^\infty dx\; xe^{-x/a}\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}xe^{-x/a}\right) = -\frac{\hbar^2}{2m}\frac{4}{a^5}\int_0^\infty dx\;(x^2-2ax)e^{-2x/a} = \frac{\hbar^2}{2m}\frac{1}{a^2}.$$

(c) We write the potential energy expectation value

$$\langle V\rangle_\psi = \frac{4F}{a^3}\int_0^\infty dx\; x^3e^{-2x/a} = \frac{4F}{a^3}6\left(\frac{a}{2}\right)^4 = \frac{3F}{2}a.$$

(d) Combining these:

$$E(a) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi = \frac{\hbar^2}{2m}\frac{1}{a^2} + \frac{3F}{2}a.$$

To find the minimum, we take the derivative with respect to a and solve for the zero of the derivative:

$$0 = \frac{d}{da}E(a) = -\frac{\hbar^2}{2m}\frac{2}{a^3} + \frac{3F}{2} \quad \Rightarrow \quad a_0 = \left(\frac{4}{3F}\frac{\hbar^2}{2m}\right)^{1/3}.$$

Substituting back into the variational energy, after some algebra we find

$$E_0 = E(a_0) = \left(\frac{3^5}{2^4}\right)^{1/3}\left(\frac{\hbar^2}{2m}F^2\right)^{1/3} \approx 2.4764\left(\frac{\hbar^2}{2m}F^2\right)^{1/3} \approx 0.8333\,\mathrm{eV}.$$

It is not necessary to perform this algebraic manipulation to get the correct result, but it shows that the eigenenergies are proportional to $\left(\frac{\hbar^2}{2m}F^2\right)^{1/3}$. This can be shown in general by rewriting the problem using dimensionless units. The problem can be solved exactly in terms of Airy functions, and the numerical constant appearing in the ground-state energy is given by the negative of the first root of the Airy function, $a_1 \approx -2.33811$, giving an exact energy about 5% lower than the variational result.